YAKEEN

NEET 2.0 2027

Physics | Lecture - 5

Motion in a Plane

By- Manish Raj (MR Sir)



Recored lecture - 4th

hr → 1hr

$$\frac{180 \text{ Days}}{90 \text{ chapt}} = 2 \text{ Days/Ps chapt}$$

Todays Goal

⇒ Question Practice on Projectile

Horizontal & Inclined Projectile Motion

H|w goal

Date - 17/08/2024

S.A → 1 (2-D]

⇒ Revision Lect 1 to 4 (2-D)

(8) Ball is projected at 50 m/s at 37° from horizontal then find time when it is moving ⊥r to initial direction of projection.

(a) 6.3 sec

(b) 8 sec

(c) 4 sec

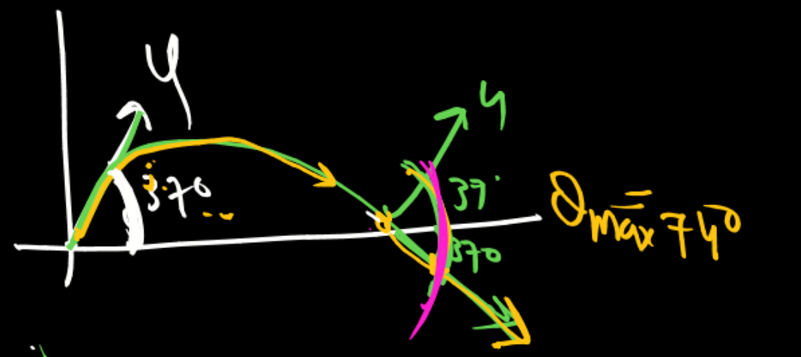
(d) 6 sec.

(e) Not Possible

$$t = \frac{u}{g\sin\theta}$$

$$= \frac{50}{10 \times \frac{3}{5}} = \frac{25}{3} = 8.3\text{ sec}$$

* If Angle of Projection is less than 45° then final velocity will become never ⊥r to initial dirn of Projection during ground to ground flight



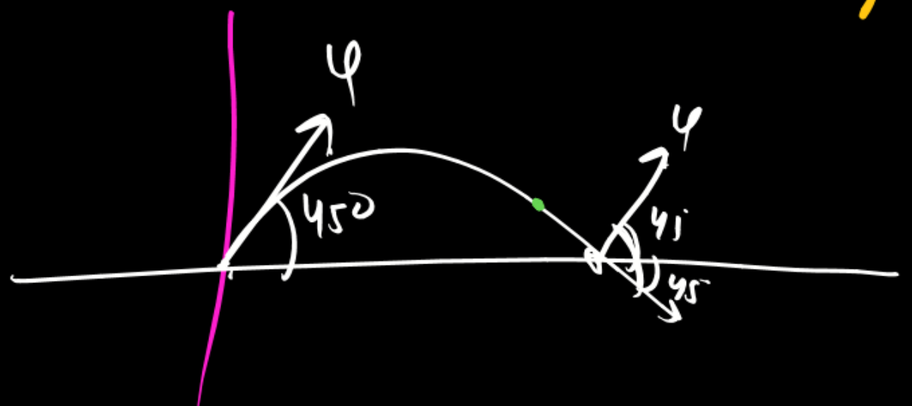
∴

copy

* In above question angle of Projection is 45°

- then time when it becomes t, $t = \frac{u}{g\sin\theta} = \frac{u}{g\sin(45)} = \frac{\sqrt{2}u}{g}$

- $T_{\text{time of flight}} = \frac{2u\sin\theta}{g} = \frac{2u\sin 45}{g} = \frac{2u}{g}\frac{1}{\sqrt{2}} = \frac{\sqrt{2}u}{g}$



(8) Ball is Projected and its position at time t is $\vec{r} = 30t\,\hat{i} + (40t - 4t^2)\,\hat{j}$ Then find Range :-

Solution

$$\vec{r} = 30t\,i + (40t - 4t^2)\,\hat{j}$$

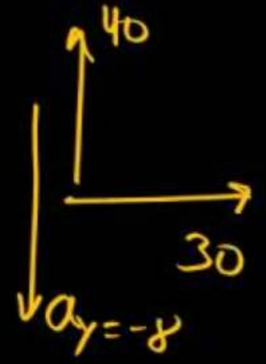
$$\vec{v} = 30i + (40 - 8t)\,\hat{j}$$

$$\vec{a} = 0 + 0 - 8 \times 1\,\hat{j}$$

$$\boxed{\vec{V}_{t=0} = 30i + 40\hat{j}}$$

$$\boxed{\vec{a} = -8\hat{j}}$$

$a_x = 0$



$$T_f = \frac{2u_y}{g} = \frac{2 \times 40}{8_4} = 10\,\text{sec}$$

MR Scan (+1)

Question

Object is moving with velocity $\vec{V} = 3\sin(\omega t)\,\hat{i} + 3\cos(\omega t)\,\hat{j}$ then find distance moved by object in 2 sec.

$\vec{V} = 3\sin(\omega t)\,\hat{i} + 3\cos(\omega t)\,\hat{j}$

→ 2-D motion

(a). acc^n = 0 (-1) pen

(b). acc$^n \neq 0$ ✓

66/.

$|\text{speed}| = \sqrt{(3\sin(\omega t))^2 + (3\cos\omega t)^2}$

dist ≠ disp^m

$\text{Speed} = 3\sqrt{\sin^2(\omega t) + \cos^2(\omega t)} = 3 \times 1 = 3\,m/s$

Speed → (cost n)

$\text{disp}^n = \text{speed} \times \text{time} = 3 \times 2 = 6m$ ✓

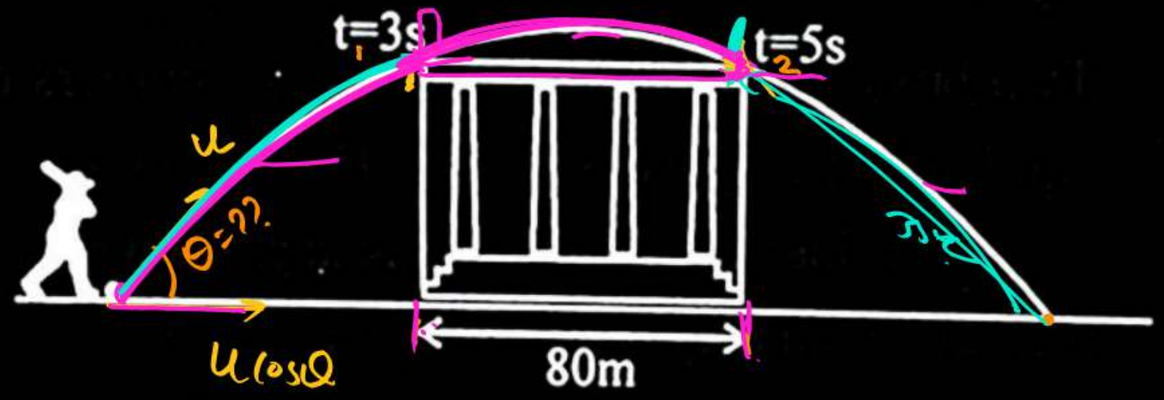
Question

The six hit by CHRIS GAYLE in IPL just misses a building of length 80 m as shown in figure. The angle of projection with horizontal is:

1. 45°
2. 30°
3. 60°
4. 15°

$\frac{u\sin\theta}{u\cos\theta} = \frac{u_0}{u_0} = 1$

$\tan\theta = 1$



$T_f = \frac{u_y}{g} = \frac{u\sin\theta}{g}$

$u_0 = u\sin\theta$

← $\Delta t = 2\,sec$ –

x-axis me math

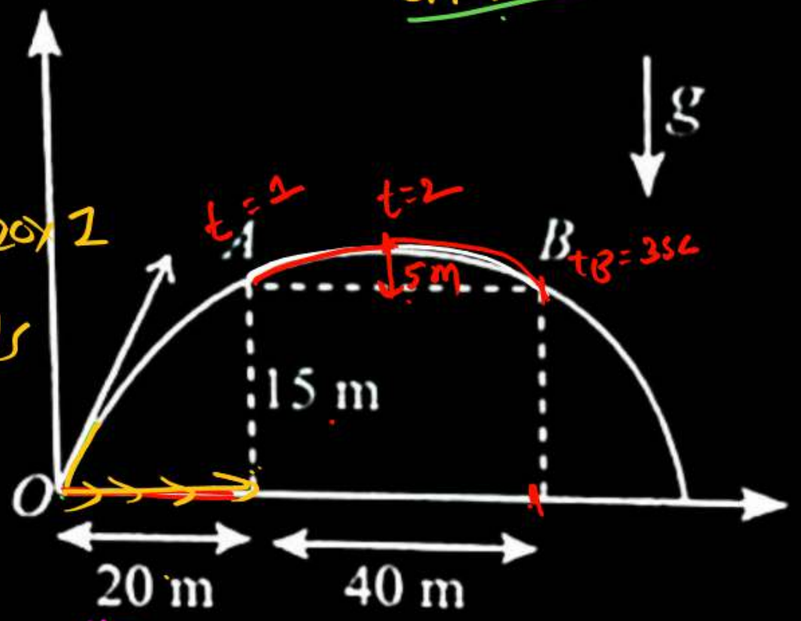
$80 = u\cos\theta \times 2$

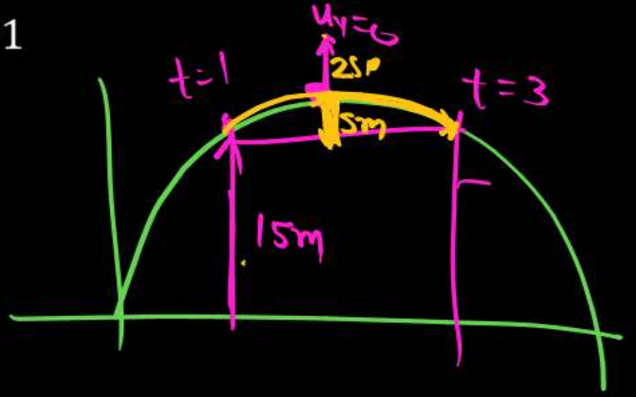
$u_0 = u\cos\theta$ — (1)

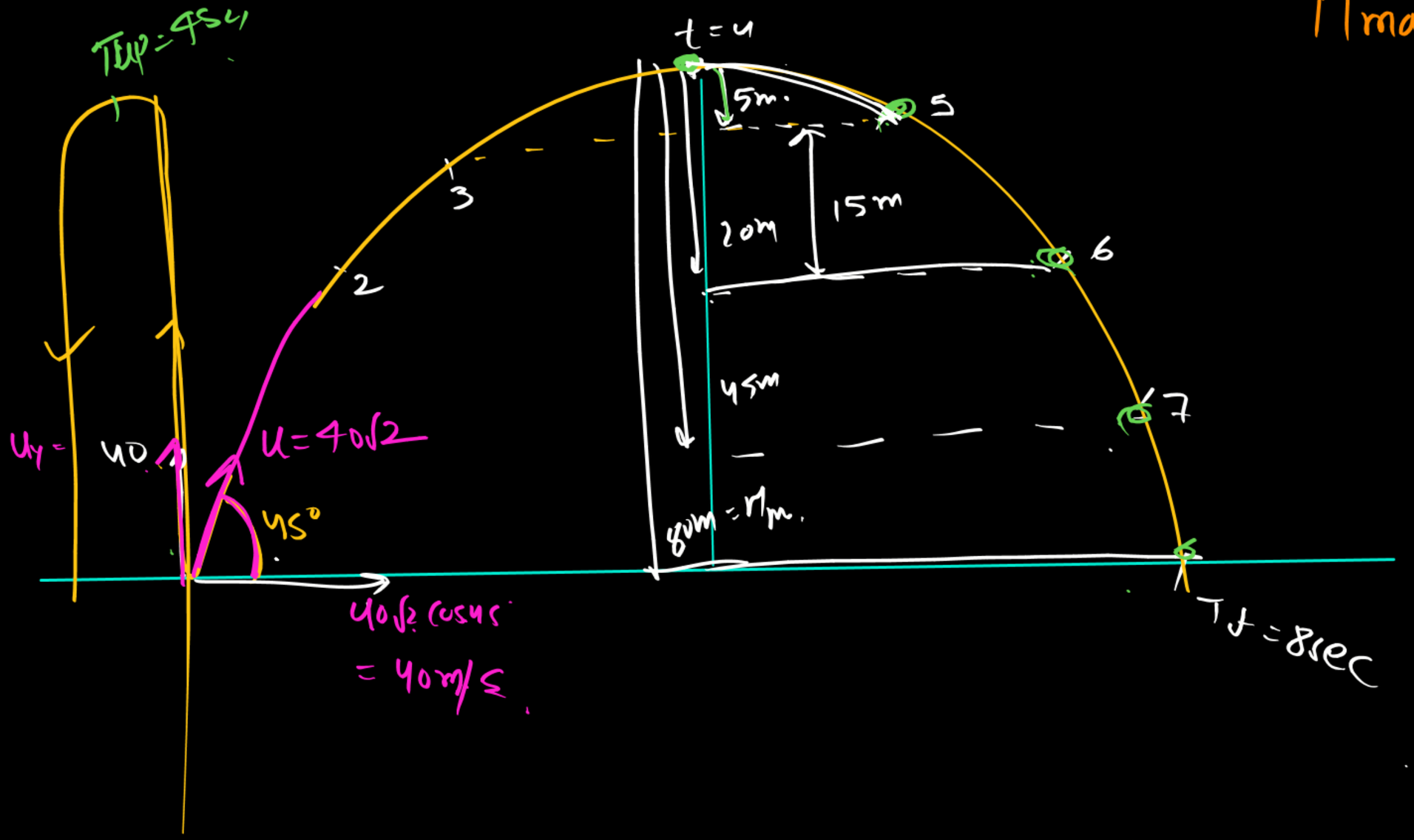
Question

In the projectile motion shown in figure, given t_{AB} = 2 sec, then (take, g = 10 ms^{-2})

1. Particle is at point 'B' at 3 sec
2. Maximum height of projectile is 20 m
3. Initial vertical component of velocity is 20 ms^{-1}
4. Horizontal component of velocity is 20 ms^{-1}







$$H_{max} = \frac{u^2 \sin^2\theta}{2g}$$

$$= \frac{(40\sqrt{2})^2 \times \sin^2 45}{2 \times 10}$$

$$= \frac{40 \times 40 \times 2 \times}{2 \times 2 \times 10}$$

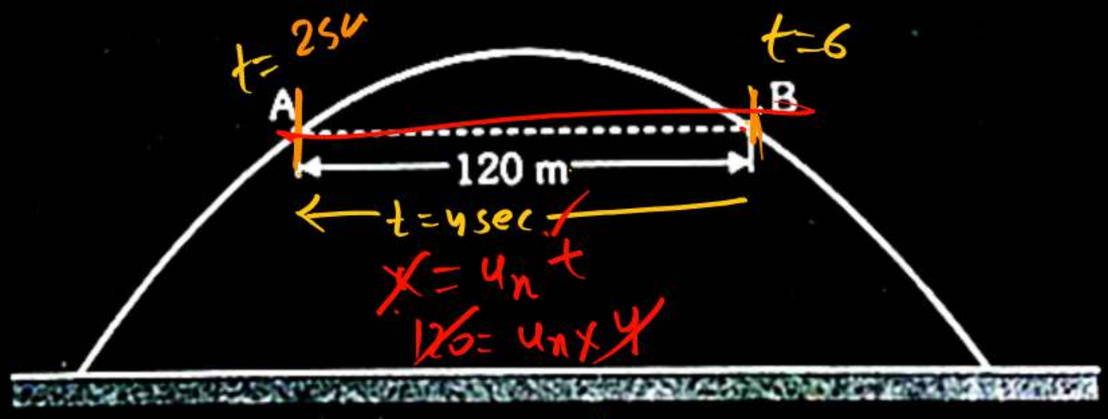
$$= 80m$$

Question

A projectile passes two points A and B at same height after 2s and 6s of its projection. Horizontal separation between the points A and B is 120 m. The horizontal range is closest to [$g = 10\ m/s^2$]

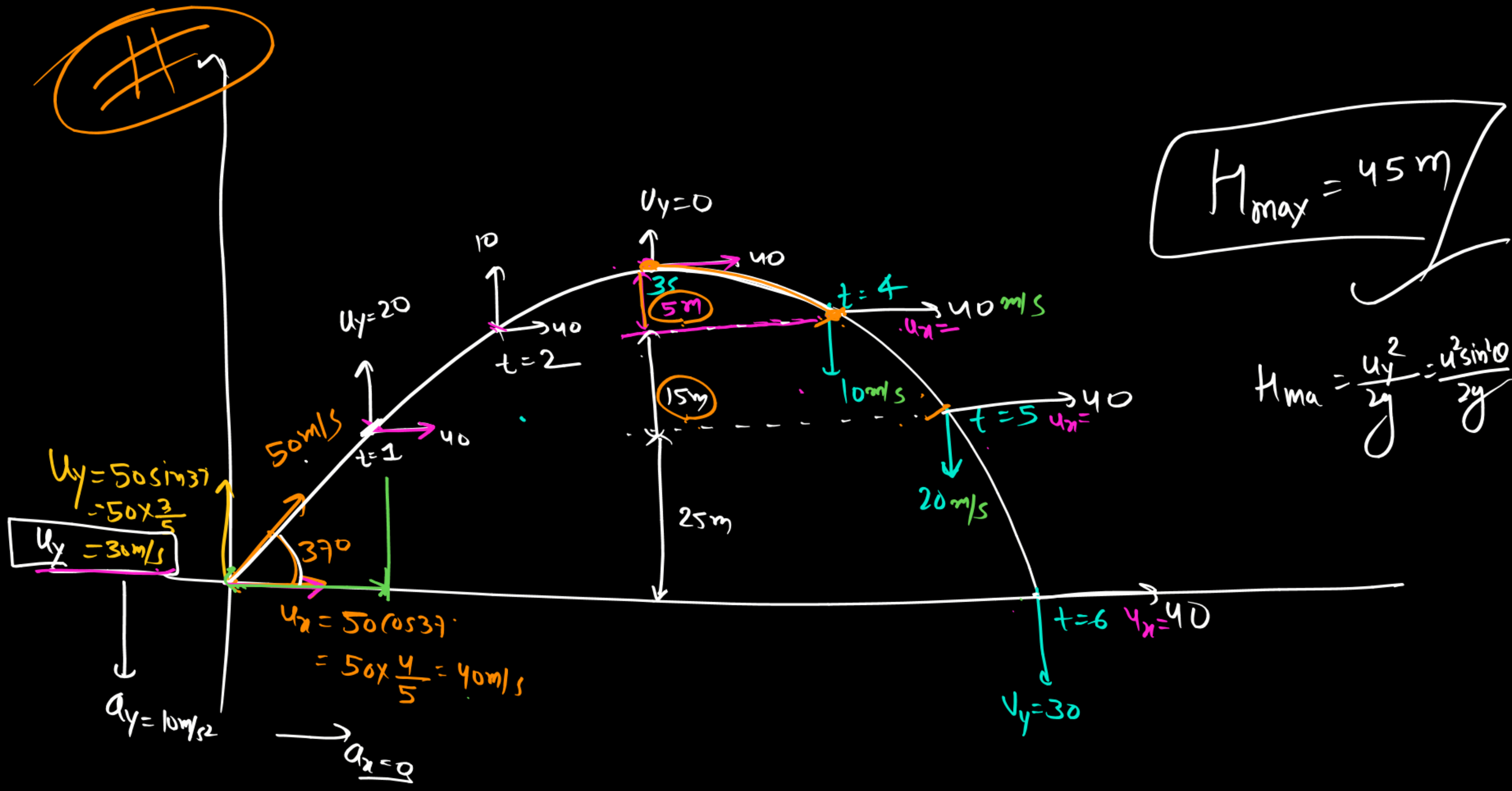
1. 180 m
2. 200 m
3. 240 m
4. 260 m

$a_x = 0$



$R = u_x T_f$

$= 30 \times 8 = 240\ m$


$H_{max} = 45\,m$
$H_{max} = \frac{u_y^2}{2g} = \frac{u^2\sin^2\theta}{2g}$
$u_y = 0$
10
40
35
5m
t = 4
40 m/s
$u_x =$
$u_y = 20$
40
t = 2
15m
10 m/s
40
t = 5 $u_x =$
50 m/s
t = 1
40
$u_y = 50\sin 37$
$= 50 \times \frac{3}{5}$
$u_y = 30$ m/s
37°
25m
20 m/s
$u_x = 50\cos 37$
$= 50 \times \frac{4}{5} = 40$ m/s
t = 6 $v_x = 40$
$v_y = 30$
$a_y = 10\,m/s^2$
$a_x = 0$

Question

The trajectory of projectile is $y = x(1 - x)$, where 'y' and 'x' are in metres. Choose the CORRECT option(s).

$y = x(1-x)$ $\quad y = x\tan\theta\left(1 - \frac{x}{R}\right)$

$\tan\theta = 1$



1. Horizontal range is 1m
2. The angle of projection is 45°
3. The horizontal range is 2m
4. The horizontal range is 30°

$\theta = 30°$

Question

Copy

A stone is projected form a horizontal plane. It attains maximum height 'H' and strikes a stationary smooth wall and falls on the ground vertically below the maximum height. Assume the collision to be elastic, the height of the point on the wall where ball will strike is:

(rebound with same speed)

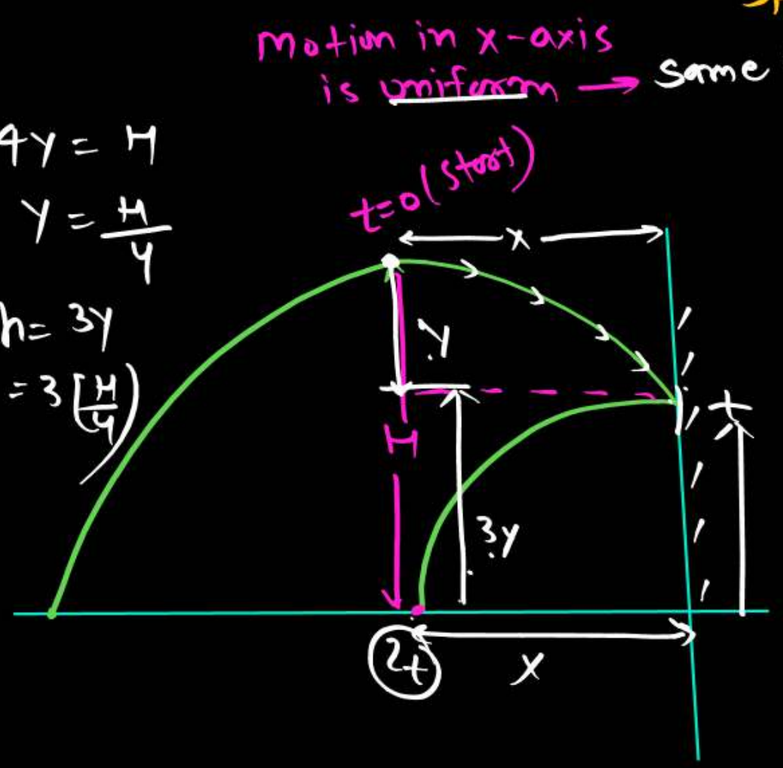
1. H/2
2. H/4
3. 3H/4 (+1)
4. None of these

Motion in x-axis is uniform → same time in same distn

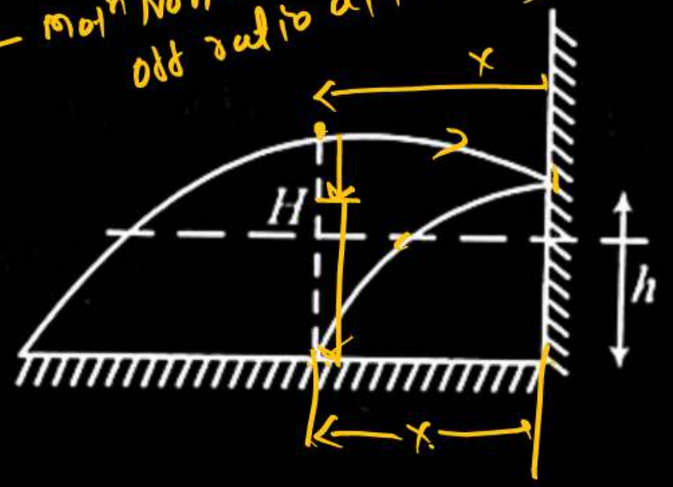
$4y = H$

$y = \frac{H}{4}$

$h = 3y = 3\left(\frac{H}{4}\right)$



y - may non-uniform odd ratio applicable

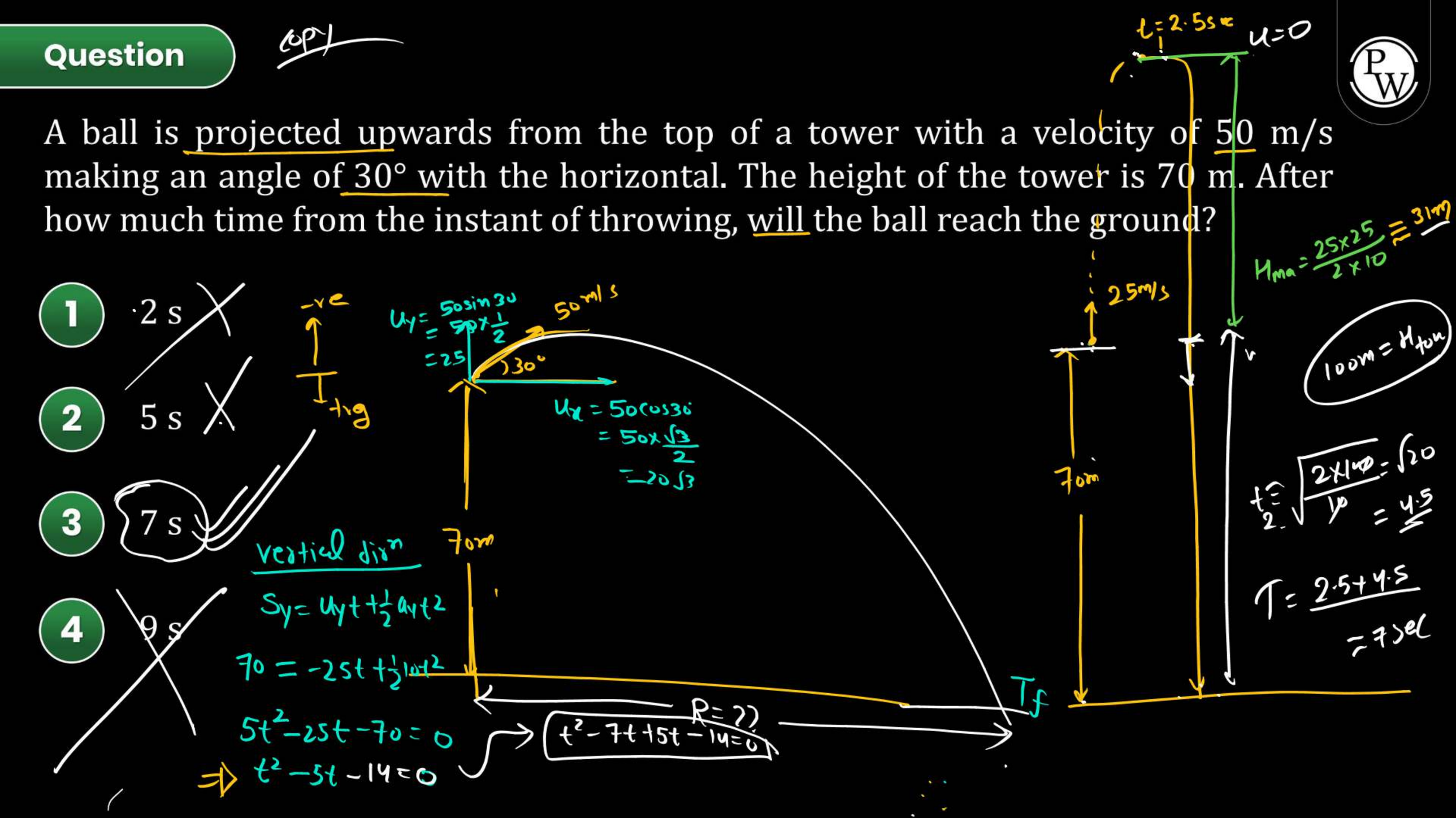


Question

copy

A ball is projected upwards from the top of a tower with a velocity of 50 m/s making an angle of 30° with the horizontal. The height of the tower is 70 m. After how much time from the instant of throwing, will the ball reach the ground?

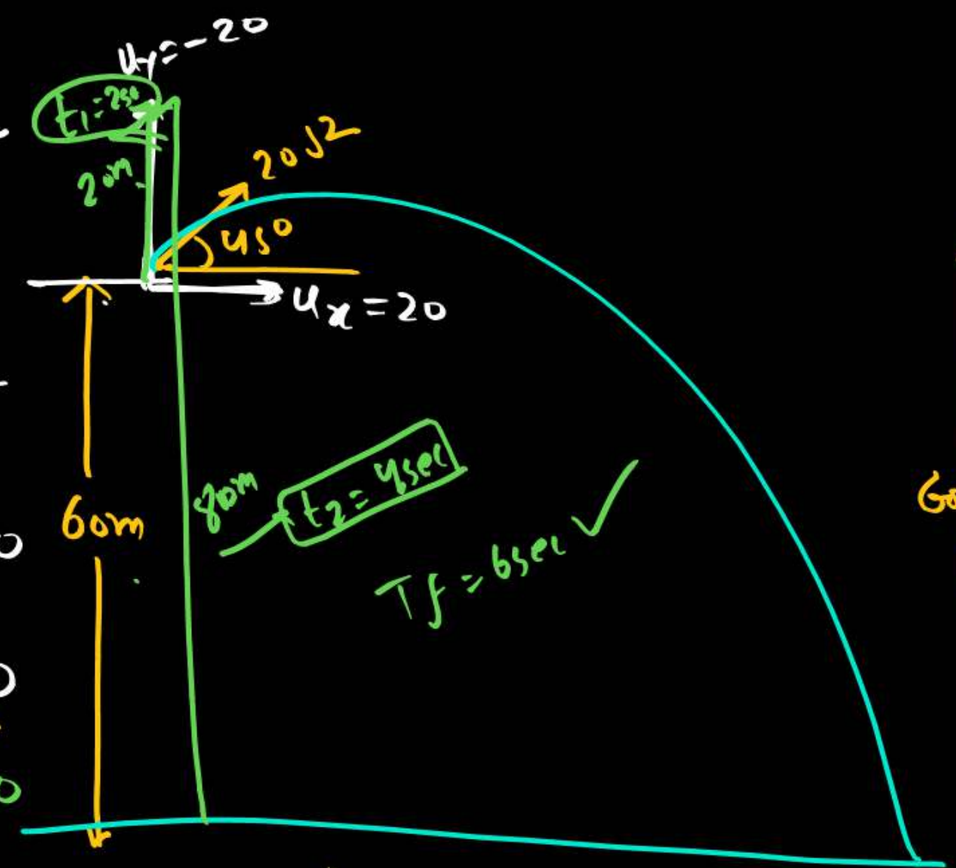
1. 2 s
2. 5 s
3. 7 s
4. 9 s

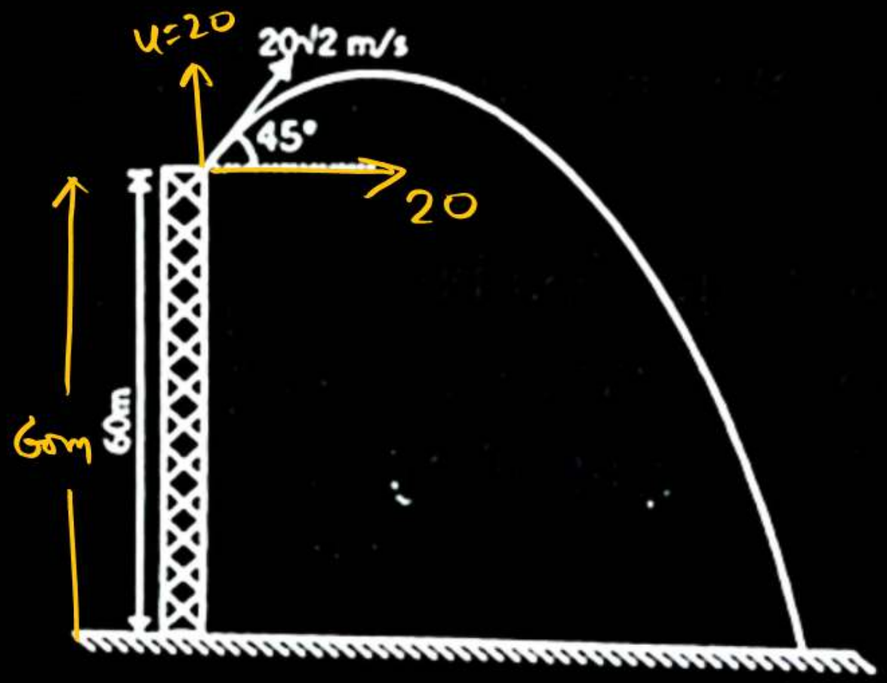


Question

A ball is thrown from the top of a 60 m high tower with velocity $20\sqrt{2}$ m/s at 45° elevation as shown in figure. F[illegible]

($g = 10\ m/s^2$)





1. Y-dirⁿ motⁿ

2. $S = ut + \frac{1}{2}at^2$

3. $60 = -20t + \frac{10}{2}t^2$

4. $5t^2 - 20t - 60 = 0$

$t^2 - 4t - 12 = 0$

$t^2 - 6t + 2t - 12 = 0$

$t(t-6) + 2(t-6)$

$t_1 = 6$ $t_2 = -2$ ✗

Question

A particle is projected from a tower as shown in figure, then the distance from the foot of the tower where it will strike the ground will be:

(take $g = 10\ m/s^2$)

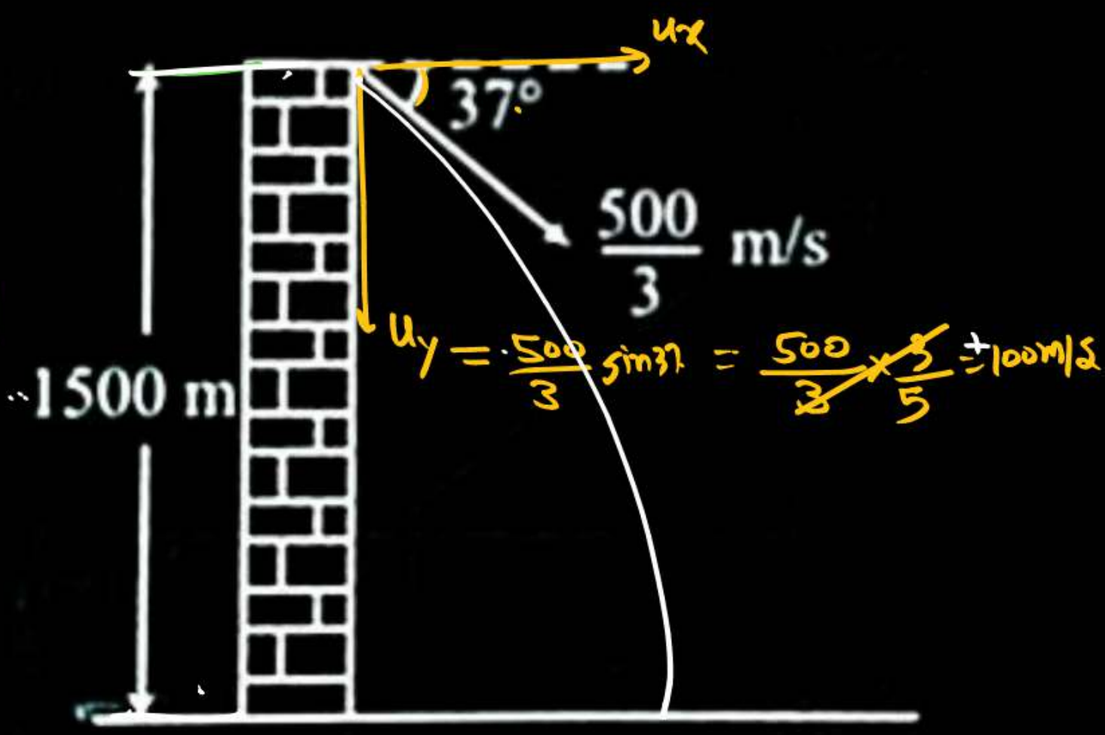
1. 4000/3 m
2. 5000/3 m
3. 2000 m
4. 3000 m

Motion in vertical dirn

$$S_y = u_y t + \frac{1}{2} a_y t^2$$

$$1500 = 100t + \frac{10}{2} t^2$$

$$5t^2 + 100t - 1500 = 0$$



Question

Aims Copy

A ball is thrown from the top of 36 m high tower with velocity 5 m/s at an angle 37° above the horizontal as shown. Its horizontal distance on the ground is closest to [$g = 10\ m/s^2$]

1. 12 m
2. 18 m
3. 24 m
4. 30 m

in y-dir^n

$S = ut + \frac{1}{2}at^2$

$36 = -3t + \frac{1}{2}10t^2$

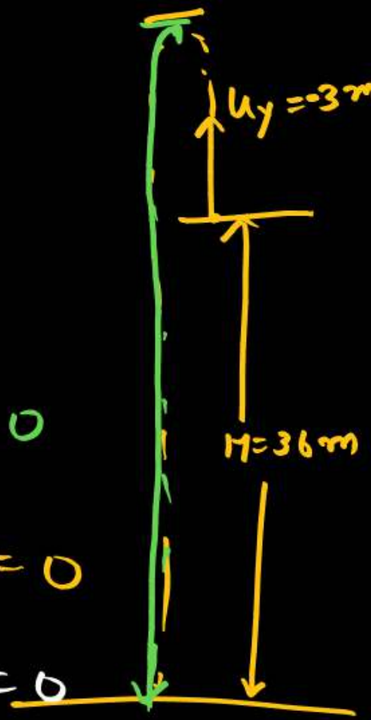
$5t^2 - 3t - 36 = 0$

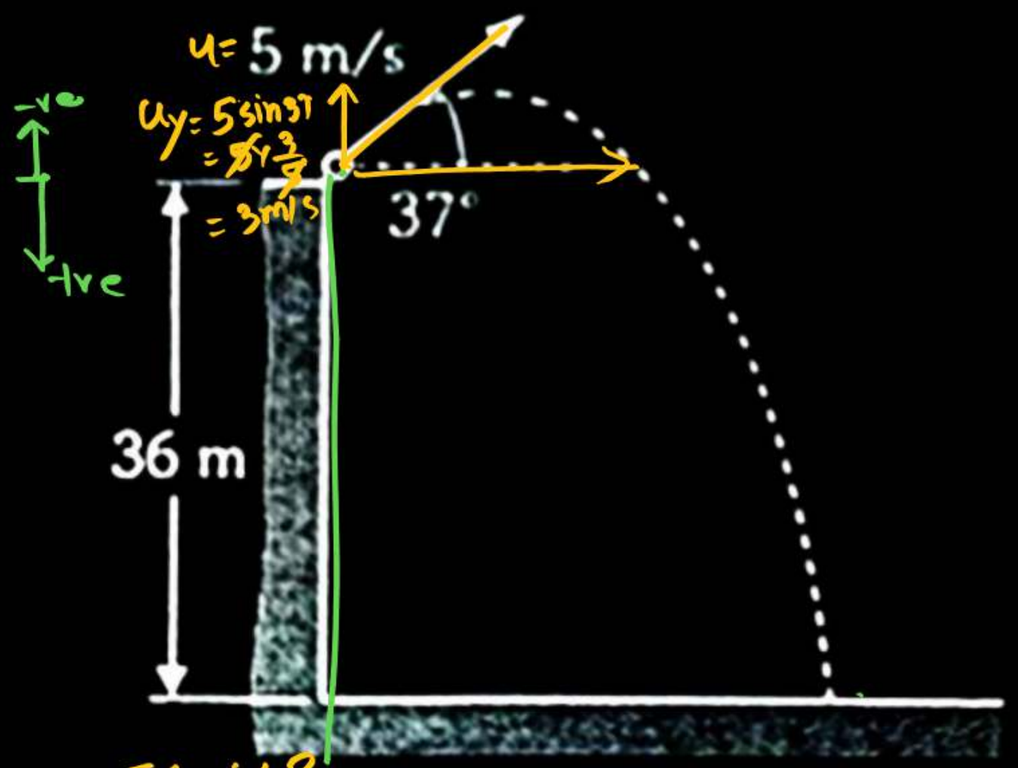
$5t^2 - 15t + 12t - 36 = 0$

$5t(t-3) + 12(t-3) = 0$

$(5t+12)(t-3) = 0$

$t_1 = 3s$ | $t = -\frac{12}{5}$ ✗



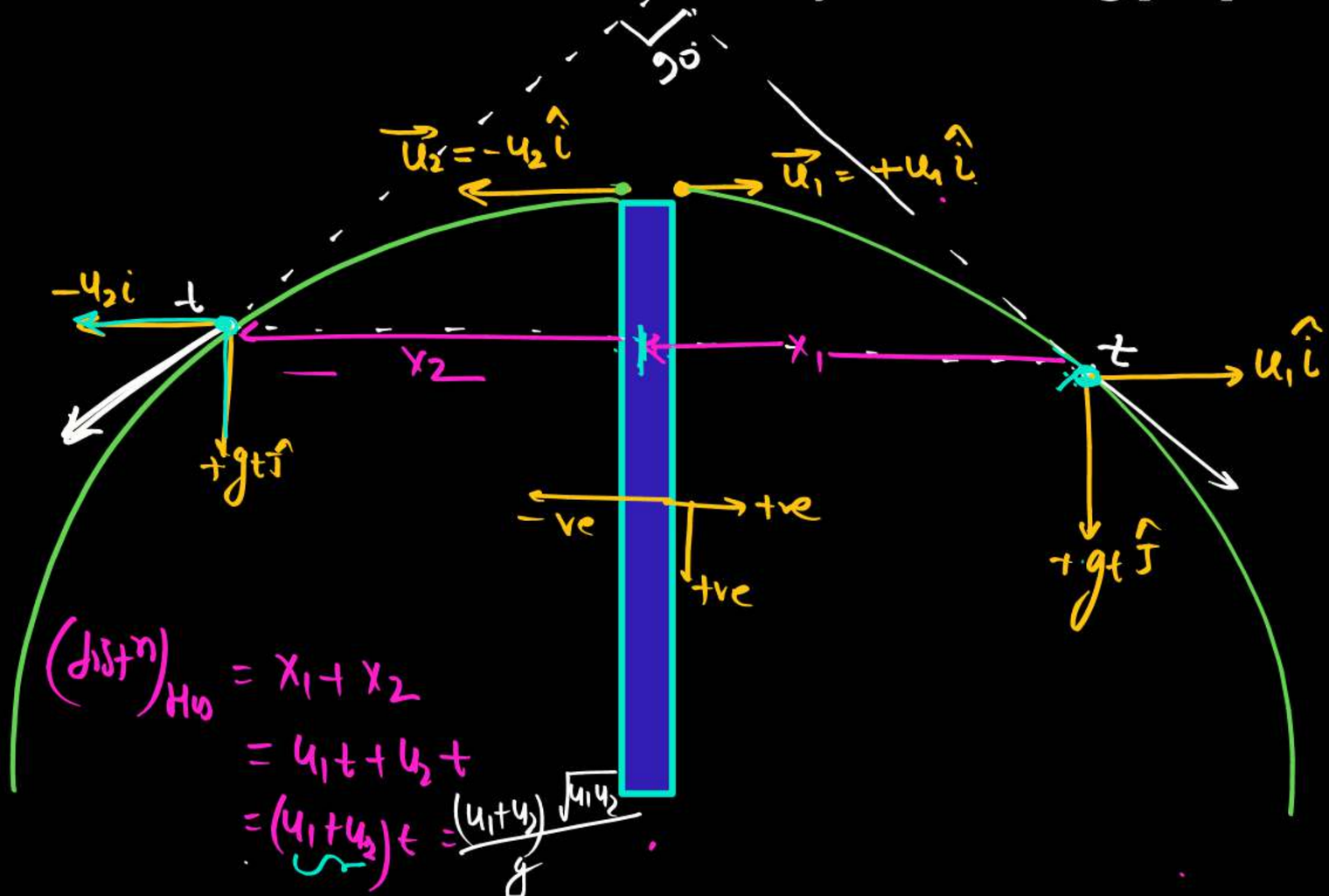


$R = u_n T_f = 5\cos 37 \times 3$

$= 5 \times \frac{4}{5} \times 3 = 12$ m,

Question

Two ball projected horizontally from height h in opposite direction with speed u_1 and u_2 then find time when they moving perpendicular to each other and also find horizontal distance between them when they are moving perpendicular.



$$\vec{V_{1t}} = u_1\hat{i} + gt\hat{j} \quad -(1)$$

$$\vec{V_{2t}} = -u_2 i + gt\hat{j} \quad -(2)$$

$$\vec{V_{1t}} \cdot \vec{V_{2t}} = 0$$

$$-u_1u_2 + (gt)^2 = 0$$

$$(gt)^2 = u_1u_2$$

$$gt = \sqrt{u_1u_2}$$

$$\boxed{t = \frac{\sqrt{u_1u_2}}{g}}$$

$$(\text{dist}^n)_{Hor} = x_1 + x_2$$

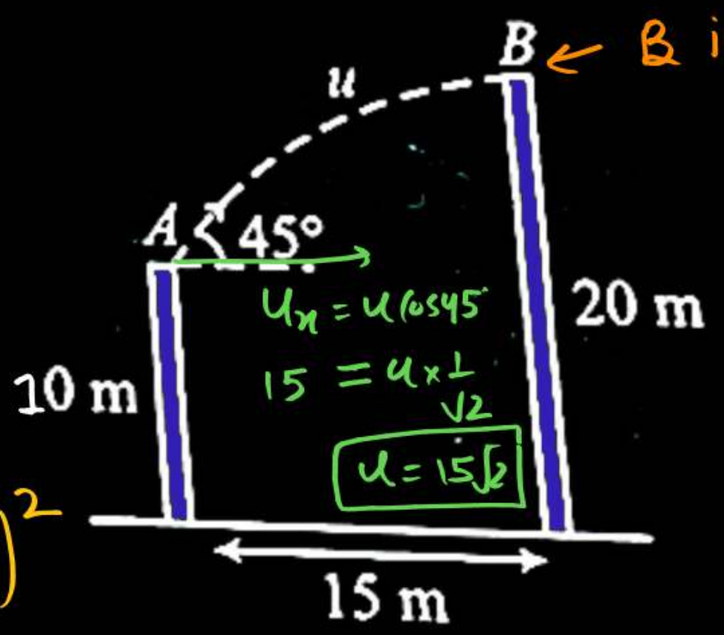
$$= u_1t + u_2t$$

$$= (u_1+u_2)t = \frac{(u_1+u_2)\sqrt{u_1u_2}}{g}$$

Question

Find the value of 'u' so that the ball reaches at point B. (Take $g = 10\ \text{m/s}^2$)

1. 20 m/s
2. 40 m/s
3. $15\sqrt{2}$ m/s
4. 50 m/s



B is not max^m height

Consider motⁿ in x-axis

$15 = u_x t_f$

motⁿ in y-axis

$10 = u_y t - \frac{1}{2} g t^2$

Put $t_f = \frac{15}{u_x}$

$10 = u_y \frac{15}{u_x} - \frac{1}{2} g \left(\frac{15}{u_x}\right)^2$

$10 = \tan\theta \times 15 - \frac{10}{2} \frac{15 \times 15}{u_x^2}$

$10 = 15 - \frac{5(15)^2}{u_x^2}$

$\frac{5(15)^2}{u_x^2} = 15 - 10$

$\frac{5(15)^2}{u_x^2} = 5$

$u_x^2 = 15^2$

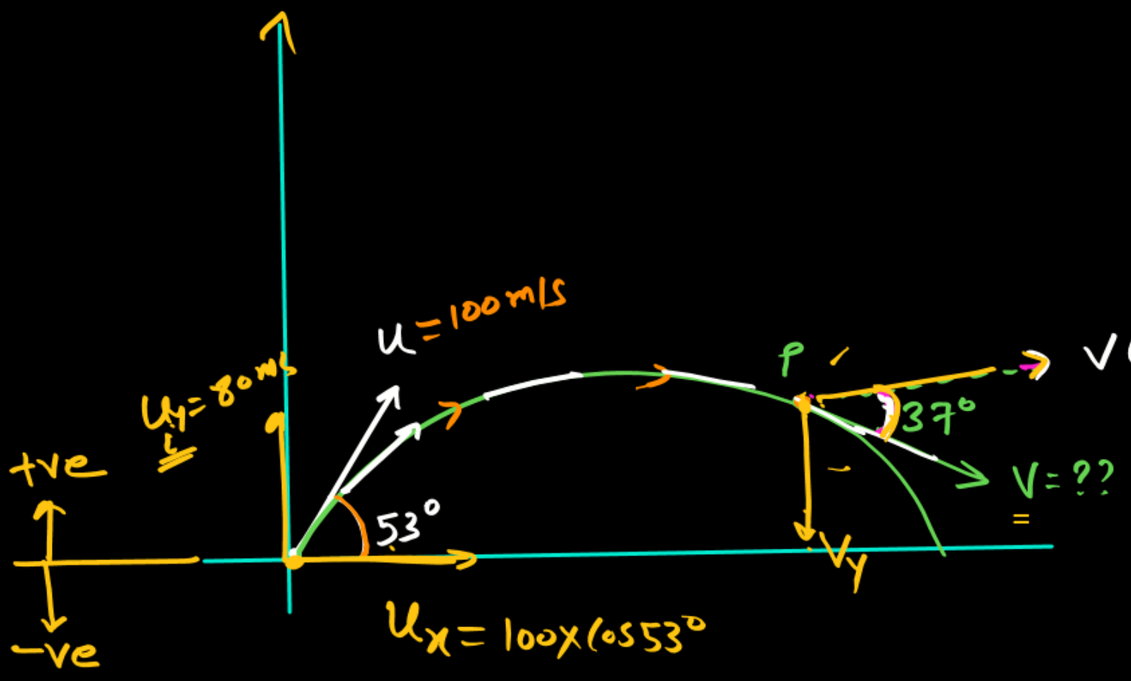
$u_x = 15$

$u_x = u\cos 45°$

$15 = u \times \frac{1}{\sqrt{2}}$

$u = 15\sqrt{2}$

find v at P and time to reach 'P'



$u_x = 100 \times \cos 53^\circ$

$= 100 \times \frac{3}{5}$

$= 60$ m/s (const^n)

Horizontal velocity (x-axis) remains constant.

$V\cos 37^\circ = u_x = 60$

$V \times \frac{4}{5} = 60$

$V = 5 \times 15 = 75$ m/s ✓

V_{speed}

at 'P' $V_y = V\sin 37^\circ = 15 \times \frac{3}{5} = 45$ m/s

$\tan 37 = \frac{V_y}{60}$

$\frac{3}{4} = \frac{V_y}{60}$

$V_y = 45$ m/s

find time to reach Point 'P'

$V_{fy} = u_{iy} + a_y t$

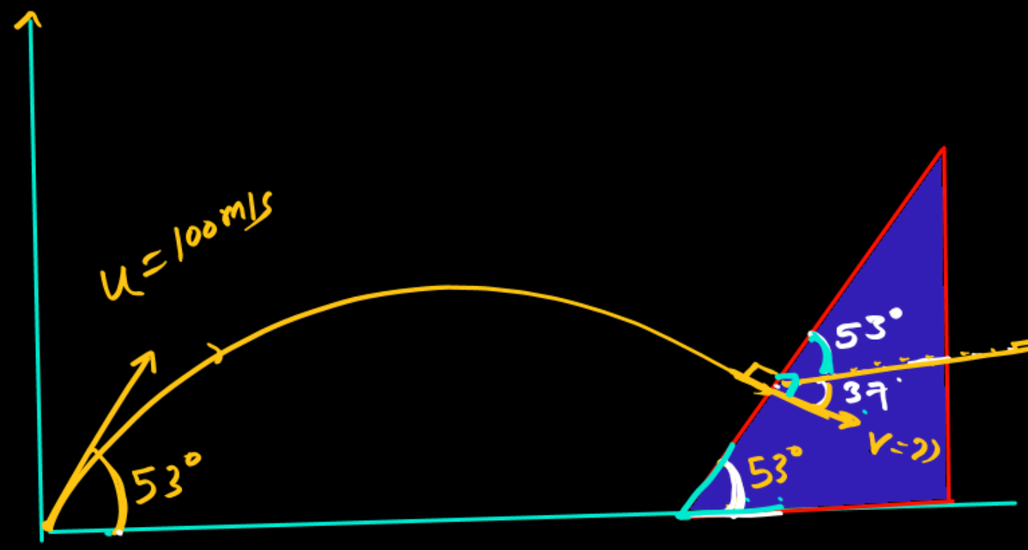
$-45 = 80 - 10t$

$10t = 125$

$t = 12.5$ sec ✓

Ball collide Perpendicular to Inclined Plane then find its speed of collision

⇒ Tf



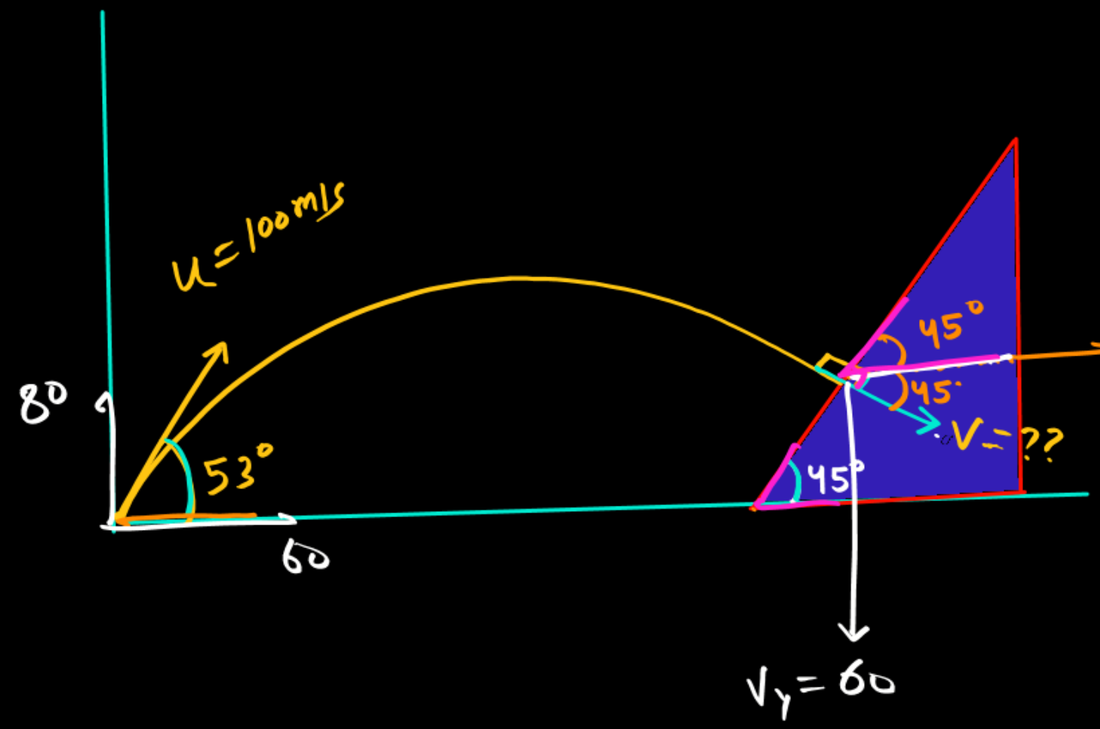
$V\cos 37^\circ = u_x = u\cos\theta$

$V\cos 37^\circ = u\cos\theta$

$V \times \frac{4}{5} = 100 \times \cos 53^\circ$

$V = 75\ m/s$

Ball collide Perpendicular to Inclined Plane then find its speed
⇒ Tf



$V\cos 45° = 100 \times \cos 53°$

$\frac{V}{\sqrt{2}} = 100 \times \frac{3}{5}$

$V = 60\sqrt{2}$

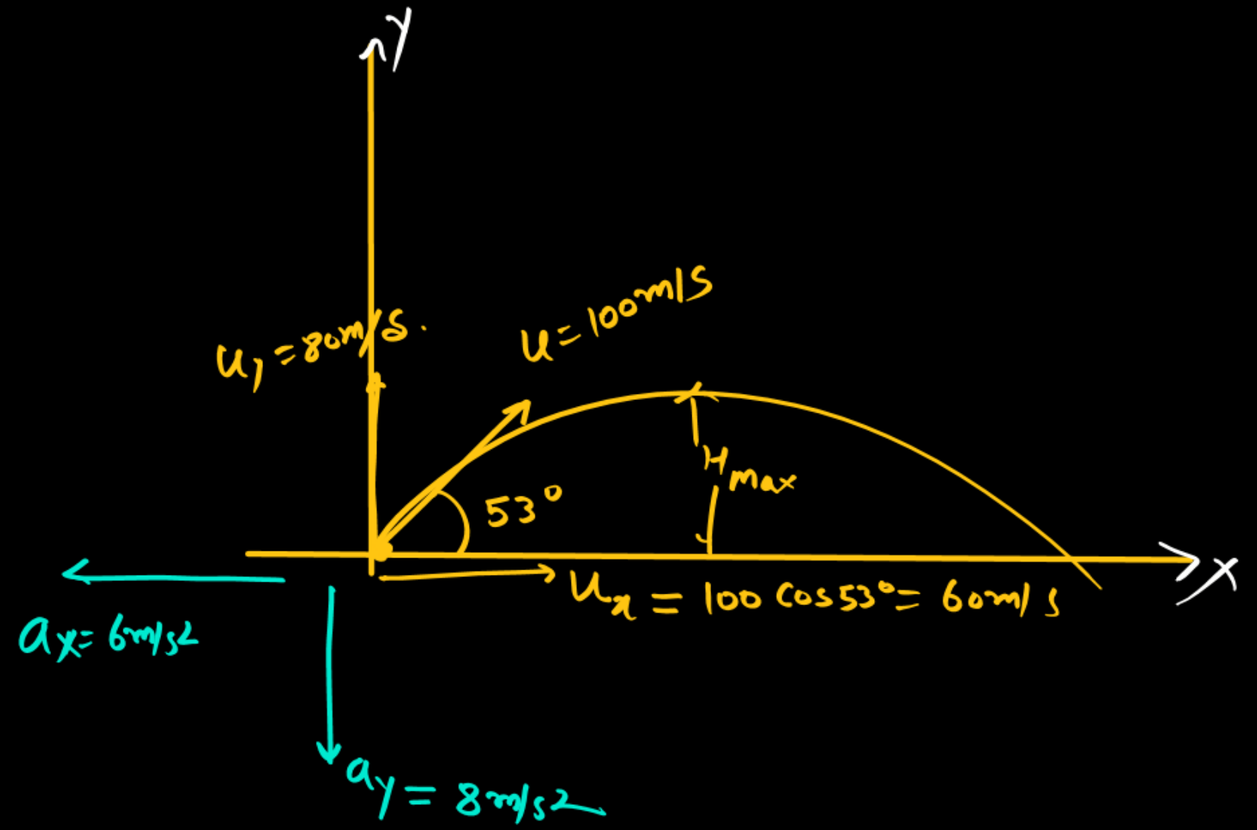
time to reach Inclin Plane

$V_f = u_y + a_y t$

$-60 = +80 - 10t$

$10t = 140$

$t = 14$ sec sec



$T_f = ??$

$H_{max} =$

$Range =$

for T_f (motion in y-axis)

$$T_f = \frac{2u_y}{(g_y)}$$

$$= \frac{2 \times 80^{10}}{8}$$

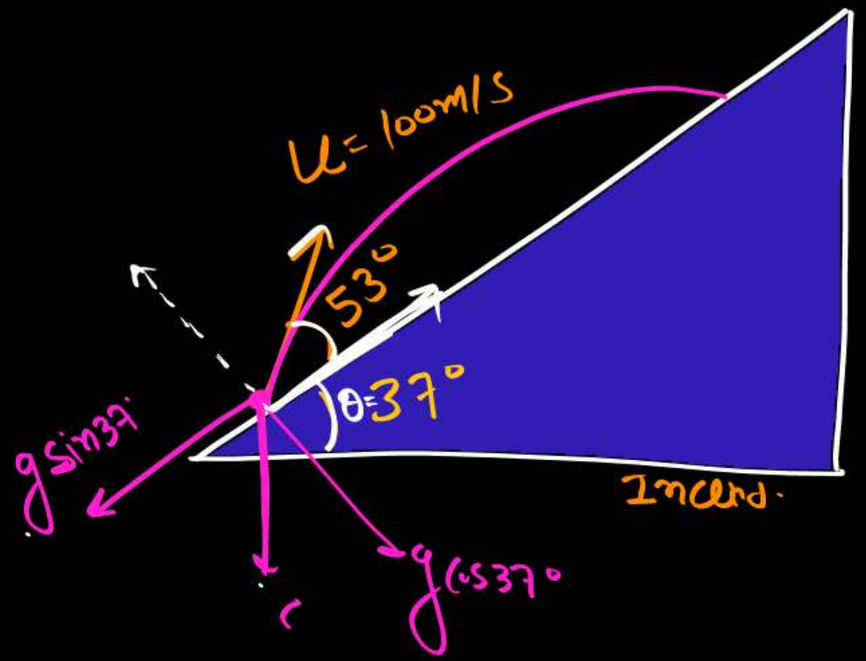
$$= 20 sec \quad \frac{5}{10}$$

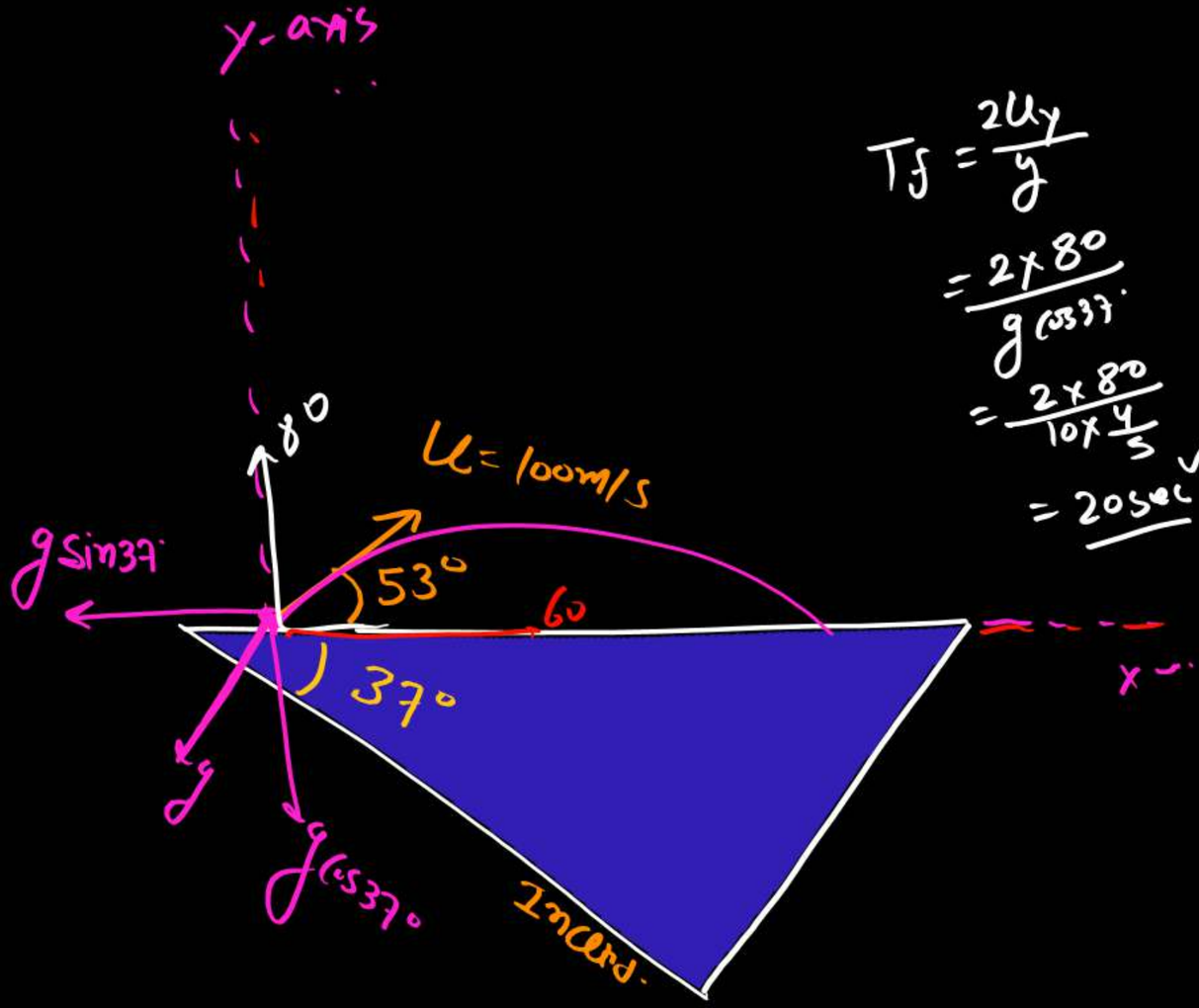
$$H_{max} = \frac{u_y^2}{2a_y} = \frac{8 \times 80^{10}}{2 \times 8}$$

$$= 400m$$

$$Rg = u_x t_f + \frac{1}{2} a_x t^2$$

Projectile motion on Inclined Plane



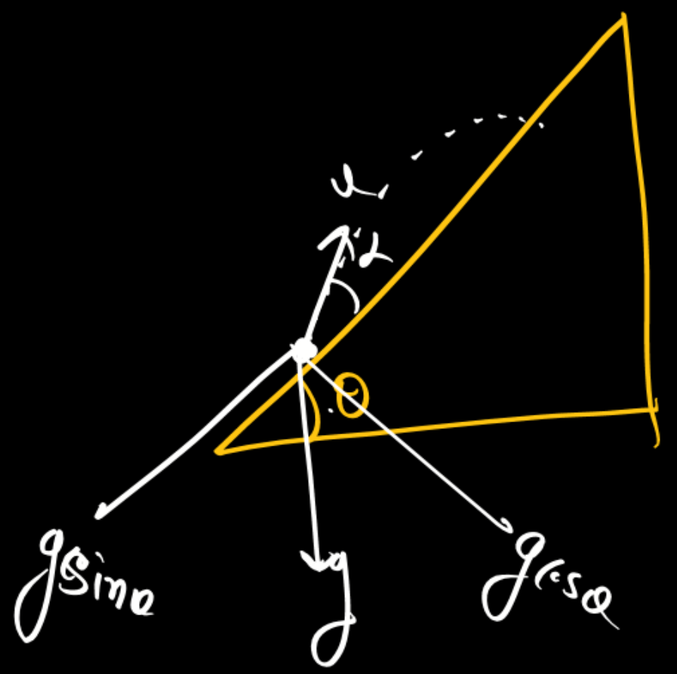


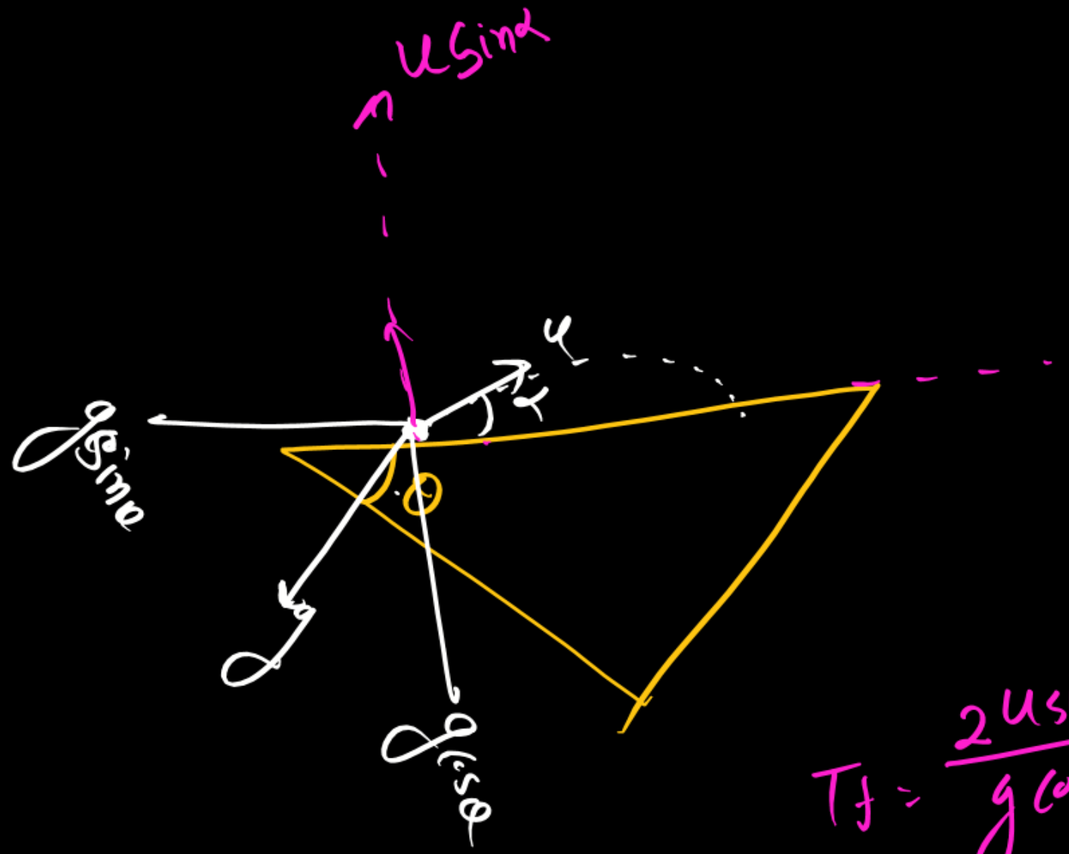
$$T_f = \frac{2u_y}{g}$$

$$= \frac{2 \times 80}{g \cos 37^\circ}$$

$$= \frac{2 \times 80}{10 \times \frac{4}{5}}$$

$$= 20 \text{ sec}$$ ✓





$$T_f = \frac{2u\sin\alpha}{g\cos\alpha}$$

Ball is projected with 100m/s at angle 37°, at maximum height Impulse imparted to the Ball so that its velocity becomes double as it's maximum height then find Tf & total Range

Question

Real 3 Fay

A projectile is thrown from a point O on the ground at an angle $45°$ from the vertical and with a speed $5\sqrt{2}$ m/s. The projectile at the highest point of its trajectory splits into two equal parts. One part falls vertically down to the ground 0.5 s after the splitting. The other part, t seconds after the splitting, falls to the ground at a distance x meters from the point O. The acceleration due to gravity $g = 10\ \text{m/s}^2$. **[JEE Adv. 2021]**

The value of t is _____.

The value of x is _____.

Question

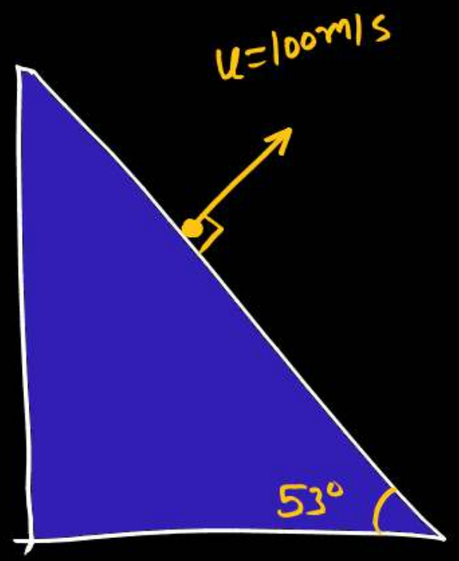
Read & Try

A projectile is fired from horizontal ground with speed v and projection angle θ. When the acceleration due to gravity is g, the range of the projectile is d. If at the highest point in its trajectory, the projectile enters a different region where the effective acceleration due to gravity is $g' = \frac{g}{0.81}$ then the new range is $d' = nd$. The value of n is _____.

[JEE Adv. 2022]

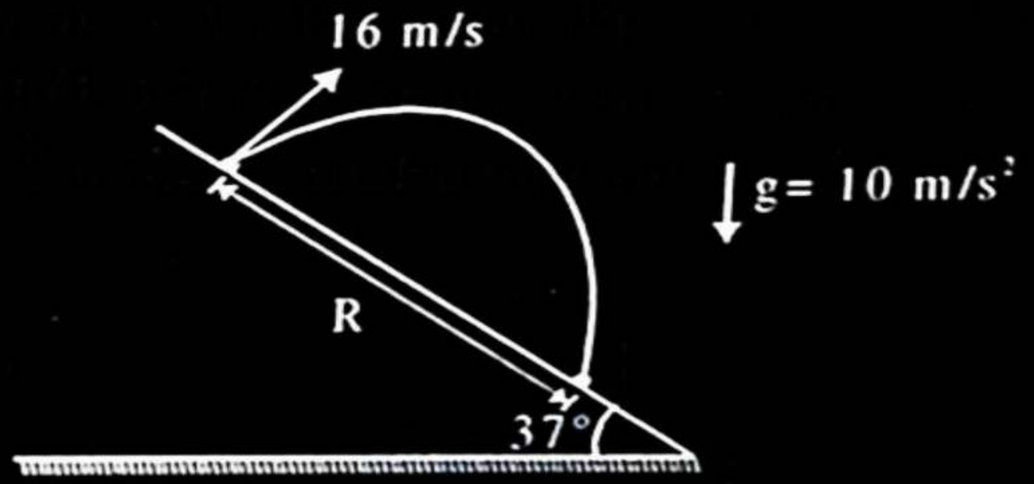
Projectile motion on Inclined Plane :—

Ball is Projected ⊥r to Inclined plane then find Tf and Range ??



Question

A projectile is launched with a velocity of 16 m/s at right angles to the slope which is inclined at 37° with the horizontal. The value of R is:

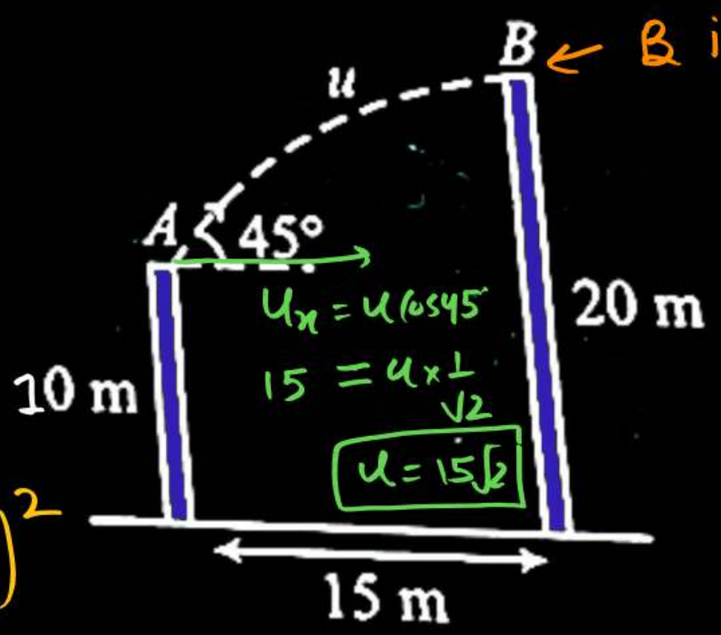


1. 96 m
2. 48 m
3. 72 m
4. None of these

Question

Find the value of 'u' so that the ball reaches at point B. (Take $g = 10\ \text{m/s}^2$)

1. 20 m/s
2. 40 m/s
3. $15\sqrt{2}$ m/s
4. 50 m/s



B is not max^m height

Consider
Motn in x-axis

$15 = u_x t_f$

Motn in y-axis

$10 = u_y t - \frac{1}{2} g t^2$

Put $t_f = \frac{15}{u_x}$

$10 = u_y \frac{15}{u_x} - \frac{1}{2} g \left(\frac{15}{u_x}\right)^2$

$10 = \tan\theta \times 15 - \frac{10}{2} \frac{15 \times 15}{u_x^2}$

$10 = 15 - \frac{5\,(15)^2}{u_x^2}$

$\frac{5(15)^2}{u_x^2} = 15 - 10$

$\frac{5(15)^2}{u_x^2} = 5$

$u_x^2 = 15^2$

$u_x = 15$

$u_x = u\cos 45°$

$15 = u \times \frac{1}{\sqrt{2}}$

$u = 15\sqrt{2}$

Thank

You